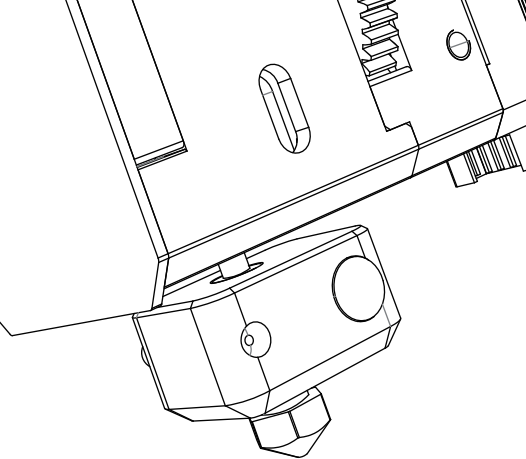
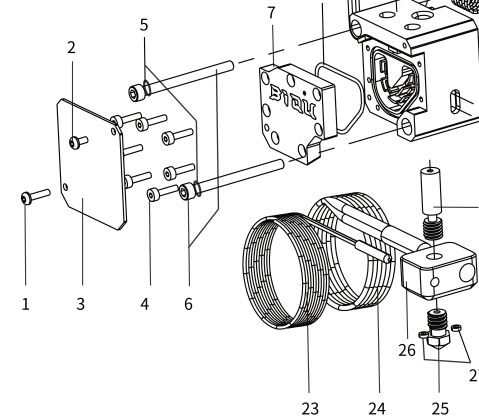




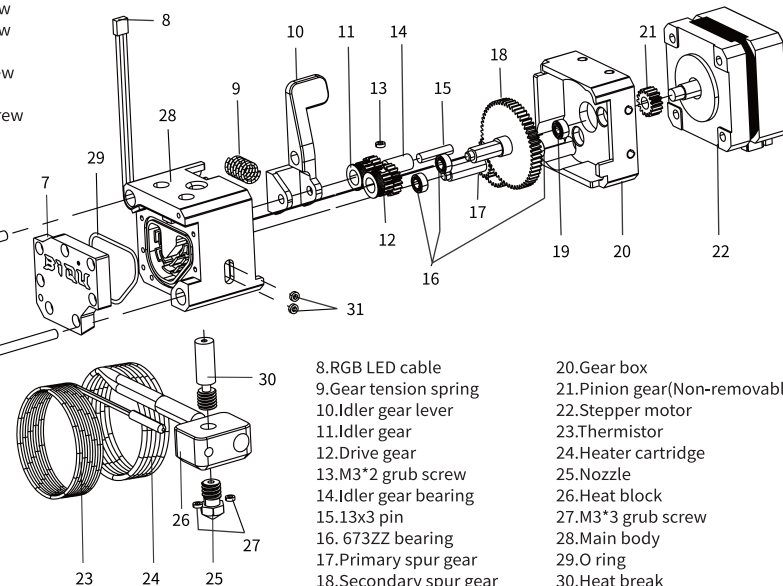
User Manual

H₂O

- 1.M2*4 round head hex screw
- 2.M2*8 round head hex screw
- 3.Cover plate
- 4.M2*6 socket head hex screw
- 5.M3 lock washer
- 6.M3*35 socket head hex screw
- 7.Water block



- 8.RGB LED cable
- 9.Gear tension spring
- 10.Idler gear lever
- 11.Idler gear
- 12.Drive gear
- 13.M3*2 grub screw
- 14.Idler gear bearing
- 15.13x3 pin
16. 673ZZ bearing
- 17.Primary spur gear
- 18.Secondary spur gear
19. 682XZZ bearing



- 20.Gear box
- 21.Pinion gear(Non-removable)
- 22.Stepper motor
- 23.Thermistor
- 24.Heater cartridge
- 25.Nozzle
- 26.Heat block
- 27.M3*3 grub screw
- 28.Main body
- 29.O ring
- 30.Heat break
- 31.M3*3 grub screw

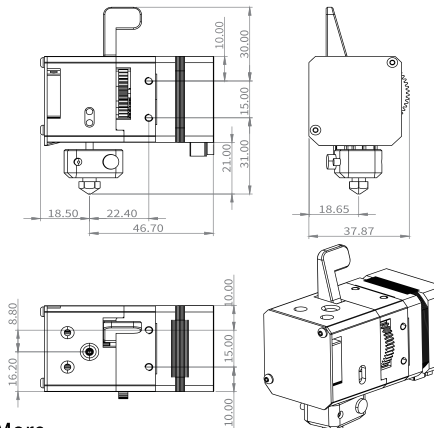
H₂O extruder manual

Summary

- It is recommended to tighten the nozzle, set screw of heat break and set screw of PT100 again when the nozzle is heated to over 400°C.
- Extrusion method: Dual gear extrusion
- Maximum printing temperature: 500°C
- Weight: 232g (Including Heat block)
- Maximum extrusion force: 7.5kg(depending on filament)
- Extrusion capacity (Based on nozzle diameter):600mm³/min (depending on filament)
- Stepper pulse: 932steps/mm @16 subdivisions(microsteps)
E-step calibrations should be performed for best accuracy
- Recommended motor current: 800mA
- Filament diameter and tolerance: 1.75±0.05mm
- Gear Ratio: 7:1
- Circumference of extrusion wheel: 24.5mm
- Extrusion wheel diameter: 7.8mm
- The maximum temperature of the silicone sleeve:
270 °C (Please remove the silicone sleeve when it is above 270 °C)
Not compatible with Nema14, but based on its design

Dimension

- XYZ size (including fan): L63.9xW37.8xH76(mm)
- Nozzle diameter: Φ 0.4mm



More

It is recommended to install the waterblock towards the left. If installing the waterblock towards the right, pay close attention to the secondary spur gear to prevent grinding on any other parts.

The eight M3 mounting holes around the gear box can be utilized for mounting the extruder and mounting accessories on to the extruder.

- The included M3x8 Screws can be used to mount parts that are no more than 5mm in thickness.

Maximum Operating Temperature

- *Note: the maximum temperature is for individual parts, not the whole system.*
- Fan: 110°C
- Motor: 180°C
- Heat break (H2 bi-metal (copper alloy + GRADE5 titanium alloy) high temperature version): 500°C
- Bearing: 120°C
- Silicone motor cable: 150°C

Thermistor Specifications

- Size: 4*30mm
- Type: PT100(2 wires)

Heating Cable Specifications:

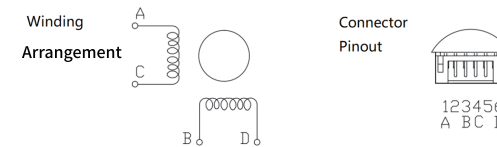
- Head size: 5*20mm
- Power: 70W

Liquid Cooling

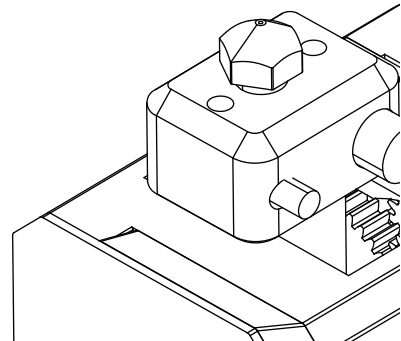
- Connector diameter: ODΦ6mm
- Coolant tube dimension: IDΦ4mm
ODΦ6mm

Motor Specifications

Itms	Specs
Rated motor cable length	1000
Rated voltage	Dc3.45V
Rated current	DC1.5A/phase
Phase number	2
Winding DC resistance(25°C)	2.3X (1±10%) Ω
Winding inductance	2.0X (1±20%) mH
Holding torque	≥110mN.m
Positioning torque	7mN.mREF
Insulation resistance	≥110MΩ (DC500V)
Insulation calss	Class H
Moment of inertia	8g.cm ³



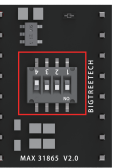
User Manual



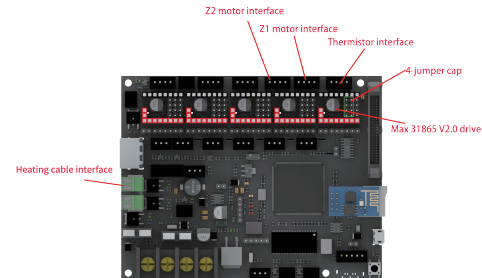
1. Hardware configuration

Driver: MAX31865 V2.0

To select PT100 mode. Select ON for 1 to 3 and OFF for 4.



Cable connection:



2. Firmware modification tutorial (Marlin bugfix-2.0.x July 07, 2021 version)

The following is an example of nozzle 0
and the same is true for nozzle 1.

(1) Configuration. h file

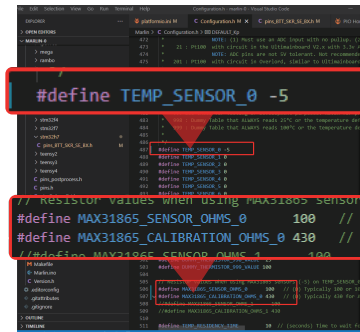
```
#define TEMP_SENSOR_0 -5
```

```
#define MAX31865_SENSOR_OHMS_0 100 // (Ω)
```

Typically 100 or 1000 (PT100 or PT1000)

```
#define MAX31865_CALIBRATION_OHMS_0 430 // (Ω)
```

Typically 430 for Adafruit PT100; 4300 for Adafruit PT1000



(2) The configuration_adv. h file

```
#define THERMOCOUPLE_MAX_ERRORS 20
```

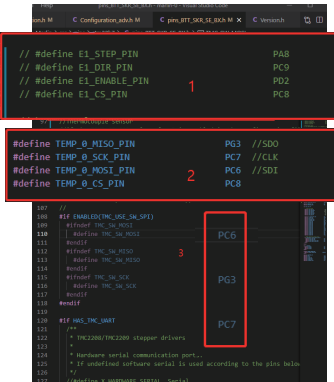
```
#define MAX_CONSECUTIVE_LOW_TEMPERATURE
```

```
_ERROR_ALLOWED 10
```

```
#define SHOW_TEMP_ADC_VALUES
```

```
#define M115_GEOMETRY_REPORT
```

(3) PIN file configuration



1) Select the driver interface , and comment out
the original driver definition to avoid
interference. The above picture takes
the BIQU BX motherboard as an example,
select the E1 driver interface to connect to
the max31865V2.0 module.

2) Define the SPI pin of the module

```
//Thermocouple sensor
```

```
//if the TEMP_SENSOR value of - 5 is enabled in the  
configuration file, it will work
```

```
#define TEMP_0_MISO_PIN PG3 //SDO
```

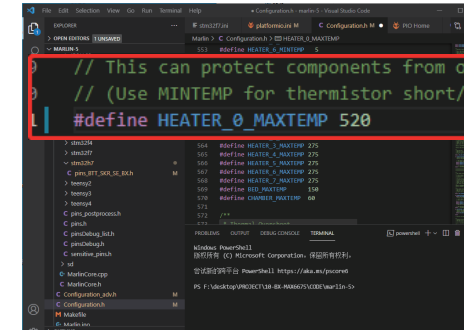
```
#define TEMP_0_SCK_PIN PC7 //CLK
```

```
#define TEMP_0_MOSI_PIN PC6 //SDI
```

```
#define TEMP_0_CS_PIN PC8
```

3) Please note that the SPI pin needs to be
consistent with the SPI pin used by the
TMC driver.

4) Modify the maximum temperature



```
#define HEATER_0_MAXTEMP 520
```

The maximum temperature exceeds 500°C